

HORNSBY GIRLS HIGH SCHOOL



2011 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question in a new booklet

Total marks (84)

- Attempt Questions 1 – 7
- All questions are of equal value

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Total Marks**Attempt Questions 1–7****All Questions are of equal value**

Begin each question in a new booklet, writing your student number and question number in the boxes indicated. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet. Marks

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{8x}{\sin 5x}$ 1
- (b) The point $C(11, -5)$ divides the interval joining $A(-3, 2)$ and B in the ratio $7:2$ internally. Find the coordinates of B . 2
- (c) Solve $\frac{2x+1}{x-3} < 3, x \neq 3$ 3
- (d) Evaluate $\int_1^9 \frac{dx}{x + \sqrt{x}}$ using the substitution $x = u^2$. 3
- (e) Find $\int (\tan x - 1)^2 dx$ 3

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Evaluate $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$. 2

(b) Consider the function $f(x) = 2 \cos^{-1}\left(\frac{x}{3}\right)$.

(i) Evaluate $f(0)$. 1

(ii) State the domain and range of $y = f(x)$. 2

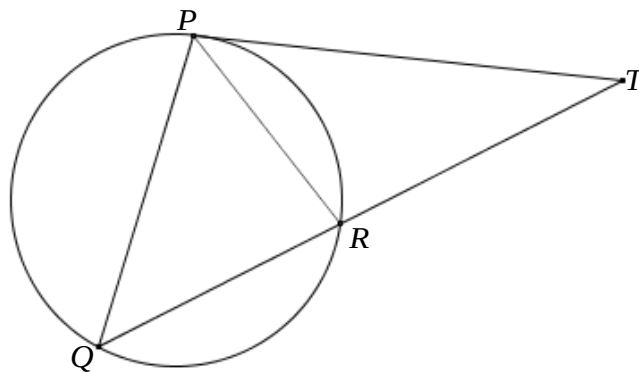
(iii) Sketch $y = f(x)$. 1

(c) A class consists of 12 girls and 10 boys.

(i) A committee of 4 is to be chosen from the class.
How many ways can this be done? 1

(ii) How many ways could the committee be chosen if it is to be made up of
3 girls and 2 boys? 2

(d) PT is a tangent to the circle PRQ and QR is a chord produced to intersect PT at T .



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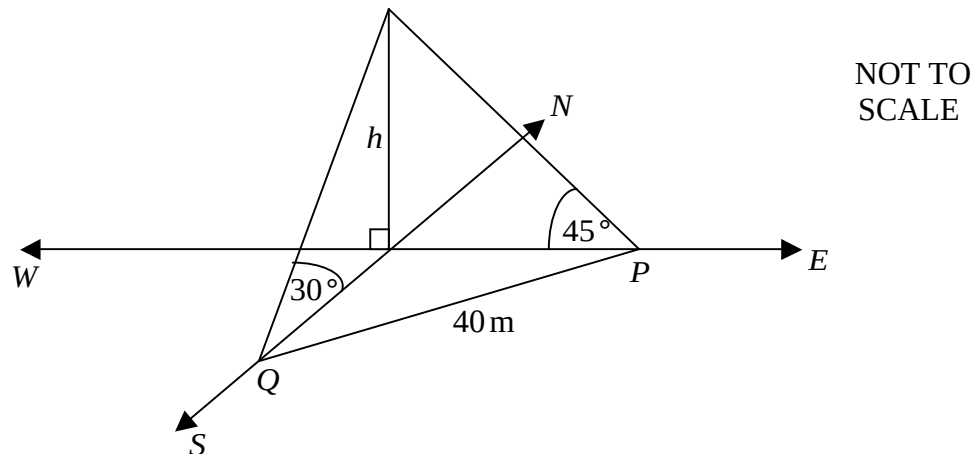
(i) Prove that $\triangle PRT$ and $\triangle QPT$ are similar. 2

(ii) Hence, prove that $PT^2 = QT \times RT$. 1

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



A vertical tower of height h metres stands on horizontal ground. From a point P , on the ground due east of the tower, the angle of elevation of the top of the tower is 45° . From a point Q , on the ground due south of the tower, the angle of elevation of the top of the tower is 30° . If the distance PQ is 40 metres, find the exact height of the tower.

3

(b) A particle P is moving along the x -axis with acceleration $\ddot{x} = -16x$, where x is the displacement of the particle from the origin. Initially, the particle is at the origin, moving with a velocity of 24 units per second.

(i) By using integration, show that the displacement is given by $x = 6 \sin 4t$, where t is time in seconds.

3

(ii) State the maximum distance from the origin that the particle reaches.

1

(iii) What is the period of the motion?

1

(iv) Sketch the graph of displacement, x , against time, t , for the first π seconds.

2

(v) Calculate the average speed of the particle during the first π seconds.

2

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Given that $x^2 + 4x + 5 \equiv (x + a)^2 + b^2$, show that $a = 2$ and $b = \pm 1$. 2

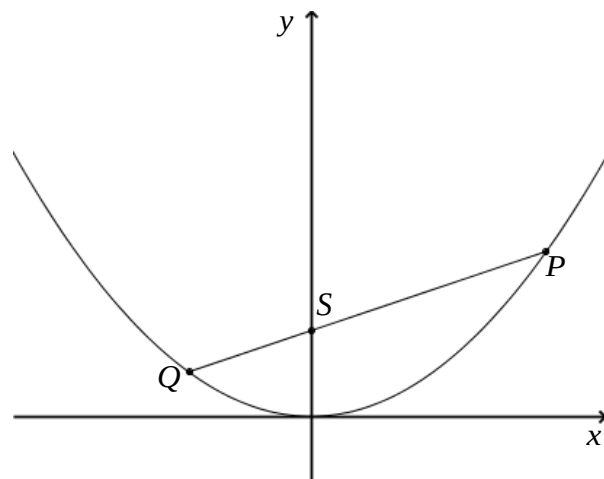
- (ii) Hence, find $\int \frac{1}{x^2 + 4x + 5} dx$. 2

- (b) At Phillips High School in NSW there are 3 Science teachers. The probability that in a NSW a Science teacher is female is 0.6. The probability that in NSW a Science teacher (male or female) is 50 years or older is 0.2.

- (i) What is the probability that at Phillips High School there is at least one female Science teacher? 2

- (ii) What is the probability that at Phillips High School all 3 Science teachers are female and younger than 50 years. 2

- (c) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$, where $a > 0$. The chord PQ passes through the focus, S .



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- (i) Show that $pq = -1$. 2

- (ii) Show that the length of chord PQ is $a \left(p + \frac{1}{p} \right)^2$. 2

Question 5 (12 marks) Use a SEPERATE writing booklet.

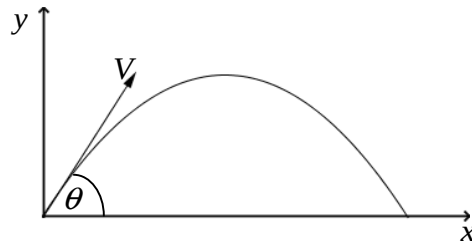
Marks

- (a) A pig farm has 100 pigs. The number of pigs, N , infected with a disease at time t days is given by $N = \frac{100}{1+ce^{-t}}$, where c is a constant.
- (i) Show that eventually all the pigs will be infected. **1**
- (ii) Initially, one pig is infected. After how many days will 70 pigs be infected? **3**
- (b) Prove by mathematical induction that $\sum_{k=1}^n k \times 2^{k-1} = 1 + (n-1)2^n$ **3**
- (c) Find the roots of the equation $x^3 - 12x^2 + 30x + 8 = 0$, given that they are consecutive terms in an arithmetic series. **3**
- (d) The population P of a country has an annual growth rate, $\frac{dP}{dt} = 0.06P$. **2**
How long will it take the population of this country to double?

Question 6 (12 marks) Use a SEPARATE writing booklet.**Marks**

- (a) A particle, P , is fired from the ground at $t = 0$. The particle is projected from the origin at an angle of θ to the horizontal, with a velocity of V . The horizontal equation of motion for the particle is

$$x_p = Vt \cos \theta. \quad \text{DO NOT PROVE THIS.}$$

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- (i) Prove that the vertical equation of motion for the particle is

2

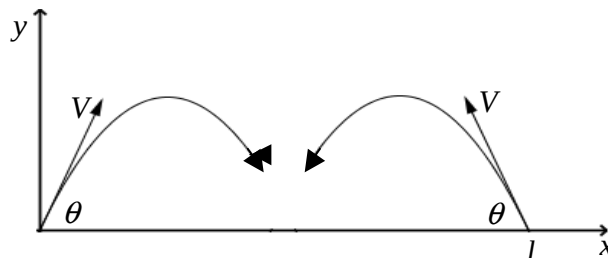
$$y_p = Vt \sin \theta - \frac{1}{2}gt^2.$$

- (ii) Show that the horizontal range of the projectile, R_p , is given by

2

$$R_p = \frac{V^2 \sin 2\theta}{g}.$$

A second particle, Q , is fired back towards the origin from the ground at a distance of l metres to the **right** of the origin at time $t = 0$, with an angle of $(180 - \theta)^\circ$ to the positive direction of the x -axis, with velocity V .

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The equations of motion for this particle are:

$$x_Q = -Vt \cos \theta + l \text{ and } y_Q = Vt \sin \theta - \frac{1}{2}gt^2. \quad \text{DO NOT PROVE THESE.}$$

- (iii) Show that if the particles collide, it will occur when $t = \frac{l}{2V \cos \theta}$.

2

- (iv) For the particles to collide, it must occur while the particles are still in flight (ie above the ground).

2

Prove that, for the particles to collide in the air, $0 < l < \frac{4v^2 \cos \theta \sin \theta}{g}$.

Question 6 continues on page 9

Question 6 (continued)

(b) Consider $f(x) = x^3 - 3x^2 - 9x$ in the domain $x \leq -1$.

(i) Find the point(s) of intersection of $y = x$ and $y = f(x)$ in this domain **2**

(ii) Hence, find the gradient of the inverse $f^{-1}(x)$ at this point. **2**

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

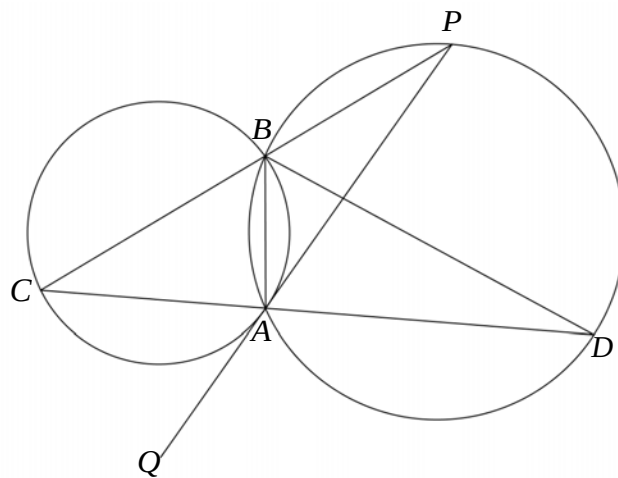
Marks

(a) It is known that $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(1-x)$ are acute angles.

(i) Show that $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$. 2

(ii) Hence or otherwise, solve the equation $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x)$. 2

(b)



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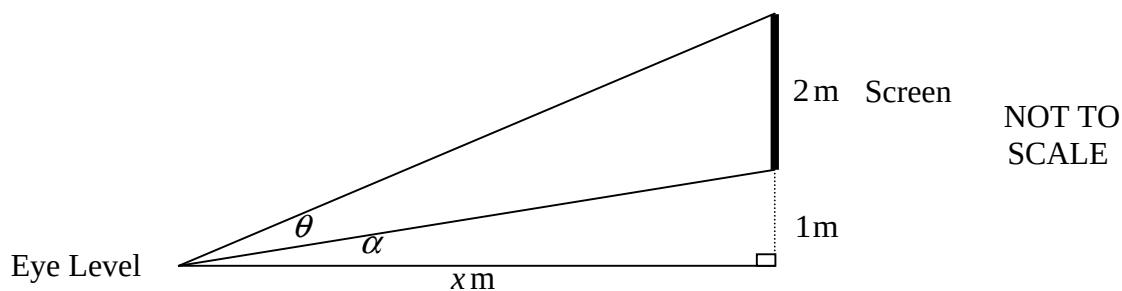
Two circles of unequal radii intersect at A and B . The tangent to the smaller circle at A cuts the larger circle at P , with PB produced cutting the smaller circle at C . The line CA produced cuts the larger circle at D .

If $\angle CAQ = \alpha$ and $\angle BAP = \beta$, show giving reasons, that $\angle ADB = \alpha - \beta$. 3

Question 7 continues on page 11

Question 7 (continued)

- (c) A projector screen on the front wall of a classroom is 2 metres high and its lower edge is 1 metre above the eye level of a seated student as indicated in the diagram. The horizontal distance of the student from the screen is x metres, the angle of elevation to the bottom of the screen is α and the viewing angle is θ . The “best” viewing angle is when θ is a maximum.



- (i) Show that $\alpha = \tan^{-1}\left(\frac{1}{x}\right)$. 1
- (ii) Show that when θ is expressed as a function of x , 2
- $$\theta = \tan^{-1}\left(\frac{2x}{3+x^2}\right).$$
- (iii) Hence or otherwise determine how far from the front of the room the student should sit in order to have the “best” view of the projector screen. 2

End of paper

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Year 12 Ext 1 HGIS Solns.

Question 1

$$a) \lim_{x \rightarrow 0} \frac{8x}{\sin 5x}$$

$$= \lim_{x \rightarrow 0} \frac{8}{5} \cdot \frac{5x}{\sin 5x}$$

$$= \frac{8}{5}$$

$$b) A(-3, 2) \quad B(x, y)$$

$$C(11, -5) \quad z = 2$$

$$\text{min}$$

$$11 = 7x + 2(-3)$$

$$9$$

$$99 = 7x - 6$$

$$7x = 105$$

$$x = 15$$

$$5 = \frac{7y + 2(2)}{9}$$

$$-45 = 7y + 4$$

$$-49 = 7y$$

$$y = -7$$

$$\therefore B(15, -7)$$

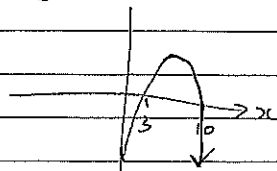
$$c) \frac{2x+1}{x-3} < 3$$

$$(2x+1)(x-3) < 3(x-3)^2$$

$$(2x+1)(x-3) - 3(x-3)^2 < 0$$

$$(x-3)[2x+1-3x+9] < 0$$

$$(x-3)(10-x) < 0$$



$$\therefore x > 10, \quad x < 3$$

$$d) F = \int_1^9 \frac{dx}{x+\sqrt{x}} \quad x=u^2$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u \cdot du$$

$$\text{When } x=1, u=1$$

$$x=9, u=3$$

$$\therefore I = \int_1^3 \frac{2u \cdot du}{u^2+u}$$

$$= \int_1^3 \frac{2}{u+1} du$$

$$= 2[\ln(u+1)]_1^3$$

$$= 2(\ln 4 - \ln 2)$$

$$= 2 \ln 2$$

$$= \ln 4$$

$$\approx 1.386 \text{ (3dp)}$$

$$e) \int (\tan x - 1)^2 dx$$

$$= \int \tan^2 x - 2 \tan x + 1 dx$$

$$= \int (\sec^2 x - 2 \tan x) dx$$

$$= \tan x + 2 \ln(\cos x) + C$$

Question 2

$$a) \int_0^{\pi/4} \cos^2 x dx$$

$$= \frac{1}{2} \int_0^{\pi/4} (1 + \cos 2x) dx$$

$$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{4} - 0 \right) - \frac{1}{2} (\sin \frac{\pi}{2} - \sin 0) \right]$$

$$= \frac{\pi}{8} + \frac{1}{4}$$

$$= \frac{1}{8} (\pi + 2)$$

$$b) f(x) = 2\cos^{-1}\left(\frac{x}{3}\right)$$

$$(i) f(0) = 2\cos^{-1} 0$$

$$= 2\left(\frac{\pi}{2}\right)$$

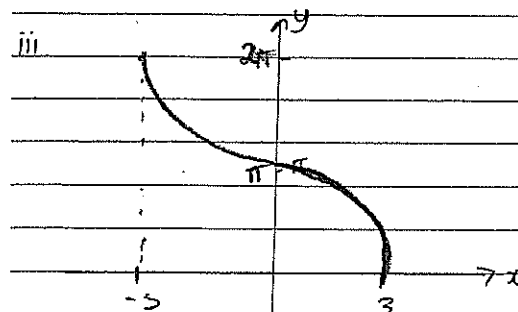
$$= \pi$$

$$(ii) D: -1 \leq \frac{x}{3} \leq 1$$

$$-3 \leq x \leq 3$$

$$R: 0 \leq \frac{y}{2} \leq \pi$$

$$0 \leq y \leq 2\pi$$



$$c) (i) {}^{22}C_4 = 7315 \text{ ways}$$

$$(ii) {}^{12}C_3 \times {}^{10}C_2 = 990 \text{ ways}$$

$$d) \text{In } \triangle PRT \text{ and } \triangle QPT,$$

$$\angle RTP = \angle QTP \text{ (common angle)}$$

$$\angle TPR = \angle TPQ \text{ (angle in alternate segment)}$$

$$\therefore \triangle PRT \sim \triangle QPT \text{ (equiangular)}$$

$$(ii) \frac{PT}{QT} = \frac{RT}{PT} \text{ (corresponding sides of } \triangle PRT \sim \triangle QPT)$$

$$PT^2 = QT \cdot RT$$

Question 3.

$$a) \tan 45 = \frac{h}{OP} \quad \tan 30 = \frac{h}{OQ}$$

$$OP = h \quad OQ = h\sqrt{3}$$

$$\therefore 40^2 = h^2 + 3h^2$$

$$h = 20$$

$$b) i) \ddot{x} = -16x$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = -16x$$

$$\frac{1}{2}v^2 = -16 \int x dx$$

$$v^2 = -16x^2 + C$$

$$\text{when } x=0, v=24$$

$$24^2 = C$$

$$C = 576$$

$$\therefore v = \sqrt{576 - 16x^2}$$

$$\frac{dx}{dt} = \frac{1}{\sqrt{576 - 16x^2}}$$

$$= \frac{1}{4\sqrt{36 - x^2}}$$

$$t = \frac{1}{4} \int \frac{1}{\sqrt{36 - x^2}} dx$$

$$t = \frac{1}{4} \sin^{-1}\left(\frac{x}{6}\right)$$

$$4t = \sin^{-1}\left(\frac{x}{6}\right)$$

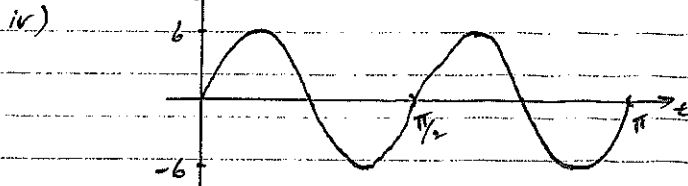
$$\frac{x}{6} = \sin 4t$$

$$x = 6 \sin 4t$$

$$v) \text{ Speed} = \frac{D}{T}$$

$$= \frac{48}{\pi}$$

$$= 15.3$$



Question 4.

$$a) i) x^2 + 4x + 4 + 1 = (x+2)^2 + 1$$

$$a=2, b^2=1$$

$$\therefore b = \pm 1$$

$$ii) \int \frac{dx}{x^2 + 4x + 5} = \int \frac{dx}{(x+2)^2 + 1}$$

$$= \tan^{-1}(x+2) + C$$

$$b) i) P(\text{at least one female}) = 1 - P(\text{none})$$

Scholar Teacher

$$= 1 - (0.4)^3$$

$$= 0.936$$

$$ii) P(3 \text{ female, c50}) = (0.6)^3 \times (0.4)^3$$

$$= 0.11 \text{ (2dp)}$$

$$c) i) \text{ Chord PQ:}$$

$$\frac{y - ap^2}{x - 2ap} = \frac{aq^2 - ap^2}{2aq - 2ap}$$

$$\frac{y - ap^2}{x - 2ap} = \frac{a(q-p)(q+p)}{2a(q-p)}$$

$$\frac{1 - p^2}{-2p} = \frac{q+p}{2}$$

$$1 - p^2 = -pq - p^2$$

$$-pq = 1$$

$$pq = -1$$

$$ii) d_{pq} = \sqrt{(2aq - 2ap)^2 + (aq^2 - ap^2)^2}$$

$$= \sqrt{4a^2(q-p)^2 + a^2(q-p)^2(q+p)^2}$$

$$= a\sqrt{4(q-p)^2 + (q-p)^2(q+p)^2}$$

$$= a\sqrt{(q-p)^2 [4 + q^2 + 2pq + p^2]}$$

$$= a\sqrt{\left(\frac{1}{p} - p\right)^2 [4 + \left(\frac{1}{p}\right)^2 - 2 + p^2]}$$

$$= a\sqrt{\left(\frac{p+1}{p}\right)^2 (p^2 + 2 + \frac{1}{p^2})}$$

$$= a\sqrt{\left(\frac{p+1}{p}\right)^4}$$

$$= a\left(\frac{p+1}{p}\right)^2$$

QUESTION 5

a) i) $\lim_{t \rightarrow \infty} e^{-t} = 0$

$\therefore \lim_{t \rightarrow \infty} \frac{100}{1+99e^{-t}} = \frac{100}{1+0}$
 $= 100$

ii) When $t=0$, $N=1$

$\therefore 1 = \frac{100}{1+c}$

$c = 99$

When $N=70$

$70 = \frac{100}{1+99e^{-t}}$

$e^{-t} = \frac{1}{231}$

$t = \ln(231)$

$= 5.44$

b) $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1) \cdot 2^n$

• Prove true for $n=1$.

L.H.S. $= 1 \times 2^0 = 1$

R.H.S. $= 1 + 0 = 1$

$\therefore$ True for $n=1$

• Assume true for $n=k+1$

i.e. $1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1) \cdot 2^n$

• Prove true for $n=k+1$

i.e. $1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) \times 2^k = 1 + k \cdot 2^k + 2^k$

L.H.S. $= 1 + (k-1) \cdot 2^k + (k+1) \times 2^k$

$= 1 + k \cdot 2^k - 2^k + k \cdot 2^k + 2^k$

$= 1 + 2k \cdot 2^k$

$= 1 + k \cdot 2^{k+1}$

$= \text{R.H.S.}$

etc.

c) Let roots be $a-d, a, a+d$

sum roots $= a-d+a+a+d = 12$

$a = 4$

prod roots $= (4-d)(4+d) \times 4 = -8$

$16-d^2 = -2$

$d^2 = 18$

$d = 3\sqrt{2}$

$\therefore$ roots are $4, 4 \pm 3\sqrt{2}$

d) $\frac{dP}{dt} = 0.06P$

$\therefore P = P_0 e^{0.06t}$

when $2P_0 = P_0 e^{0.06t}$

$2 = e^{0.06t}$

$t = \frac{\ln 2}{0.06}$

Question 6

a) (i) $\ddot{y} = -g$

$\dot{y} = -gt + c$

when $t=0$

$\dot{y} = V \sin \theta$

$V \sin \theta = c$

$\therefore \dot{y} = -gt + V \sin \theta$

$y = -gt^2 + V \sin \theta t + d$

when $t=0, y=0$

$\therefore d=0$

$\therefore y = -gt^2 + V \sin \theta t$

ii) let $y=0$

$0 = -gt^2 + V \sin \theta t$

$0 = -t(gt - V \sin \theta)$

$\therefore gt = V \sin \theta \quad (t \neq 0)$

$t = \frac{2V \sin \theta}{g}$

Sub into x

$x = \frac{V}{g} (2V \sin \theta) \cdot \cos \theta$

$= \frac{2V^2 \sin \theta \cos \theta}{g}$

$= \frac{V^2 \sin 2\theta}{g}$

ii) Part a have same y -values at time t . Need $x_a = x_p$ at some time.

$l - V \cos \theta = V \cos \theta$

$l = 2V \cos \theta$

$t = \frac{l}{2V \cos \theta}$

(iv) [There are a few methods] time at flight is $\frac{2V \sin \theta}{g}$.

Need collision time to be less than this.

$0 < \frac{l}{2V \cos \theta} < \frac{2V \sin \theta}{g}$

$0 < l < 4V^2 \sin \theta \cos \theta$

[$l > 0$, since l is to right of origin].

(b) $f(x) = x^3 - 3x^2 - 9x$

let $x = x^3 - 3x^2 - 9x$

$0 = x^3 - 3x^2 - 10x$

$0 = x(x^2 - 3x - 10)$

$0 = x(x-5)(x+2)$

$\therefore x=0, x=-2, x=5$

But $x \leq -1$

$\therefore$ Pt of intersection is $(-2, 15)$

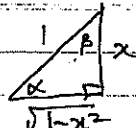
(ii) $f(x) = 3x^2 - 6x - 9$

$f'(-2) = 3(-2)^2 - 6(-2) - 9$

$= 15$

$\therefore f'(x)$ has gradient $\frac{1}{15}$ at this point.

Q7a) let $\sin^{-1} x = \alpha$
 $\cos^{-1} x = \beta$



i) $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$
 $= x \cdot x - \sqrt{1-x^2} \cdot \sqrt{1-x^2}$
 $= x^2 - (1-x^2)$
 $= 2x^2 - 1$

ii) $\sin(\sin^{-1} x - \cos^{-1} x) = \sin(\sin^{-1}(1-x))$
 $2x^2 - 1 = 1-x$
 $\therefore 2x^2 + x - 2 = 0$
 $x = \frac{-1 \pm \sqrt{1+4(2)(-2)}}{2(2)}$

$\therefore x = \frac{-1 \pm \sqrt{17}}{4} \quad (x = \frac{-1+\sqrt{17}}{4}, x > 0)$

b) $\angle CAQ = \alpha$
 $= \angle ABC$ (Angle in alternate segment)
 $\angle BAP = \beta$
 $= \angle BCA$ (Angle in alternate segment)
 $\angle APB = \angle ADB$ (Angles on same arc AB)
 $\angle ABC = \angle BAP + \angle APB$ (exterior $\angle$ of $\triangle ABP$)
 $= \angle BAP + \angle ADB$
 $\therefore \angle ADB = \angle ABC - \angle BAP = \alpha - \beta$

c) i) $\tan \alpha = \frac{1}{x}$
 $\therefore \alpha = \tan^{-1}(\frac{1}{x})$

ii) $\tan(\theta + \alpha) = \frac{3}{x}$
 $\frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = \frac{3}{x}$

$\tan \alpha = \frac{1}{x} \therefore x(\tan \theta + \frac{1}{x}) = 3 - \frac{3}{x} \tan \theta$
 $x^2 \tan \theta + 3 \tan \theta = 2x$

$\tan \theta = \frac{2x}{x^2+3}$

$\therefore \theta = \tan^{-1}(\frac{2x}{x^2+3})$

Q7iii) $\frac{dy}{dx} = \frac{1}{1 + (\frac{2x}{3+x^2})^2} \times \frac{2(3+x^2) - 2x \cdot 2x}{(3+x^2)^2}$

$= 0$ when $6 + 2x^2 - 4x^2 = 0$

ie $6 - 2x^2 = 0$

$x = \pm \sqrt{3}$

$\therefore x = \sqrt{3}, x > 0$